



PRIVAPALS: GAMIFYING DIGITAL PRIVACY EDUCATION FOR CHILDREN

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PROBLEM STATEMENT

Children ages 9 to 13 are among the most surveilled demographics on the internet, yet few resources prepare them to navigate that reality.





GAPS IN PRIVACY EDUCATION



Existing educational tools fail to teach digital literacy in an engaging way that also translates into actionable, real-world skills.

ISSUE #1: LACK OF ACCESSIBILITY

- Not all schools offer privacy education.
- Varies by state, discipline, and age group; often only available to older students, not kids 9-13.
- Limits learning to the classroom.

ISSUE #2: POOR PEDAGOGY

- Curriculums are unengaging.
- Focuses on "don'ts" and rule following.
- Difficult to understand harm in context (e.g., Failing to visualize invisible privacy risks).

ISSUE #3: PRIVACY IS COMPLEX

- Not one definition of privacy; it varies by situation and context.
- Abstract concepts are difficult for children to understand.
- Real-world privacy risks & consequences are often invisible.

OUR SOLUTION

A free-to-play game that teaches elementary to middle school children about online risks and navigating everyday privacy risks.



SI 332: FIRST PITCH OF PRIVAPALS





BEYOND SI 332: GOAL OF PRIVAPALS



After playing, kids should be able to...

- (1) Identify at least two ways apps collect their data without consent,*
- (2) Understand and explain privacy-related harms,*
- (3) Know and deploy actionable strategies to protect their personal information without adult assistance.*

PRIVAPALS: RESEARCH & EDUCATIONAL DEVELOPMENT



Defining Privacy for Kids: Privacy is context-dependent, but privacy always relates back to protecting players in digital environments.



Gameplay Structure: Scaffolded learning: Entry-level mission covering basics; unlock advanced missions based on privacy literacy to give players autonomy.



Ensuring Educational Development: Check-ins throughout and after missions; confirm concepts are understood before progressing.



Reward & Feedback Mechanics: Privacy coins earned through understanding. Privacy Shield that grows with their digital literacy, helps users visualize privacy risks & vulnerabilities.



Target Group for Prototype: Ages 9 to 10 (4th grade).

PRIVAPALS: TECHNICAL DEVELOPEMT

HAUNTED DATA MANSION

Live Demo: graceashworth.com/Privapals

- Interactive browser-based game with 3D graphics.
- Privacy threats = interactive ghosts (trackers, phishing, ads, malware).
- Check-ins to measure players awareness of privacy concepts.





PRIVAPALS: GAME DESIGN

Game Missions

←

Document tabs +

Signup Guardian — FU... ⋮

Password Potions — FULL P...

Friend or Foe — FULL PROD...

The Tracking Web — FULL P...

Data Detective — FULL PRO...

Scam Sweep — NOT DONE

Signup Guardian — FULL PRODUCTION SPEC

CORE DESIGN

In this game, the player is NOT entering their own information.

Instead:

- The player is given a predefined character profile
- The player has access to **more** information than necessary
- The player must decide what to use vs what to withhold

This creates a structured decision space where:

- All inputs are known
- All outcomes are predictable
- Tradeoffs are intentional

The goal is to simulate realistic decision-making under conditions of excess information.

PLAYER OBJECTIVE

The player's objective is to successfully create an account for their assigned character by selecting only the information that is necessary while avoiding oversharing sensitive personal details. The player must complete the signup process while minimizing exposure, understanding that each piece of information contributes to both immediate system feedback and longer-term consequences.

PLAYER DECISION FRAMEWORK

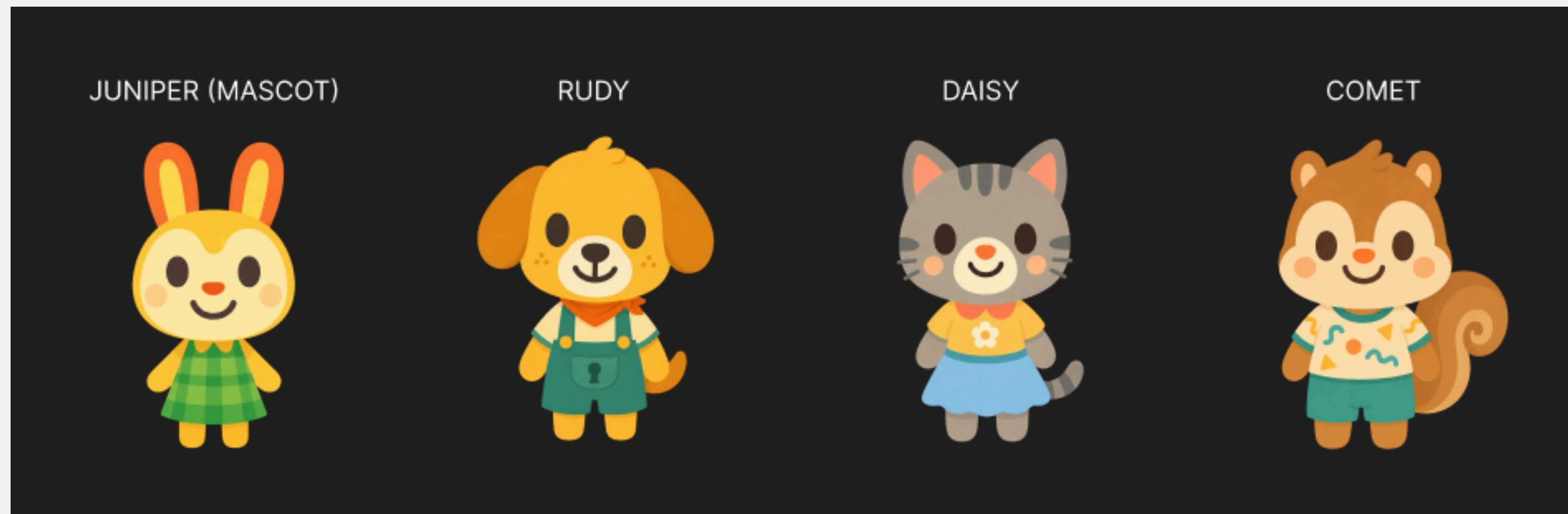
All decisions in this game follow the same underlying structure:

Player Selects Information → System Evaluates Sensitivity → System Assigns Risk Value → System Updates Privacy State → System Triggers Feedback → System Stores Consequence for Later Reveal

This loop occurs repeatedly during the form interaction.

PRIVAPALS: GAME DESIGN

Character Development + Why This Approach



Age Appropriate, Minimal, Easy to Animate, Distinctive



PRIVAPALS: GAME DESIGN



Use of AI

- Would like to minimize use of AI
- BIG QUESTION: Can we do the animation without it?
- Whats Working: Combining hand drawing with AI



PRIVAPALS: GAME DESIGN



Copyright

- Some fonts are free for PERSONAL USE
- Music is royalty free
- Art is ours completely



PRIVAPALS: GAME DESIGN



Music + Drawings + Animation



PRIVAPALS: GAME DESIGN



This Screen Becomes...

LOADING SCREEN WIREFRAME

Loading Screen
Rotating Photos of Character Art
Music Here

Tips on Privacy protection in the corner



**Strong passwords mix letters,
numbers, and symbols.**

LOADING...



PRIVAPALS: GAME DESIGN



Things We Need to Do

- Onboard Interns
- Complete Wireframing
- Complete Art Frames
- Figure Out Animation
- Write all Scripts
- Complete Hi-Fis
- Voiceovers

THANK YOU!
QUESTIONS?

